Let $p$ be a prime number.

Lemma. $\Phi_p(x) = \dfrac{x^p-1}{x-1} = x^{p-1} + x^{p-2} + \cdots + x + 1$

is irreducible.

Proof. observe that $\Phi_p(x+1) = \left[(x+1)^p - 1\right]/x = \dfrac{1}{x}\cdot\left(x^p + \binom{p}{1}x^{p-1} + \cdots + \binom{p}{p-1}x + \binom{p}{0} - 1\right)$

$$= x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + p,$$

so applying the eisenstein Criterion, we obtain the result.

Lemma. Let $\zeta_p \overset{\text{def}}{=} \exp(2\pi i \cdot \frac{1}{p})$. Then, the set

$$\left\{\zeta_p^0, \zeta_p^1, \cdots, \zeta_p^{p-1}\right\} \cong \mathbb{Z}_p.$$

Moreover, $\mathbb{Q}(\zeta_p)$ is the splitting field of $x^p-1$ and also $\Phi_p$, with $[\mathbb{Q}(\zeta_p):\mathbb{Q}] = p-1$.

Proof. It is trivial, since $p$ is a prime number. ☐

Lemma. $\left[\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}\right] = \dfrac{p-1}{2}$.

Proof. Since $\zeta_p^{-1} = \zeta_p^{p-1} = \exp(2\pi i \cdot \frac{p-1}{p})$, $\zeta_p + \zeta_p^{-1} = 2\cdot\cos\left(2\pi \cdot \frac{1}{p}\right) \in \mathbb{R}$

Then, $x^2 - (\zeta_p + \zeta_p^{-1})x + 1 \in \mathbb{Q}(\zeta_p + \zeta_p^{-1})[x]$ and it has roots as $\zeta_p$ and $\zeta_p^{-1}$.

For odd prime $p$ $(p \geq 3)$, $\zeta_p \notin \mathbb{Q}(\zeta_p + \zeta_p^{-1})$, so the polynomial above is irreducible.

Therefore, $\left[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\zeta_p + \zeta_p^{-1})\right] = 2$. Now, the tower law gives the result.

(If $p = 2$, then $\zeta_2 = -1$, so proved.)

Thm. Splitting field of $x^p - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$, and $\left[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}\right] = p(p-1)$,

Proof. Since for each $0 \leq i \leq p-1$, $\left(\sqrt[p]{2}\cdot\zeta_p^i\right)^p = 2\cdot\zeta_p^{pi} = 2\cdot 1 = 2$,

so the splitting field of $x^p - 2$ is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$. To evaluate the degree, observe that

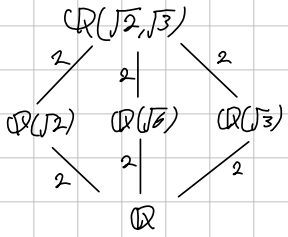
$$\mathbb{Q}(\sqrt[p]{2}, \zeta_p) = \mathbb{Q}(\sqrt[p]{2})(\mathbb{Q}(\zeta_p)$$ and

$\left[\mathbb{Q}(\sqrt[p]{2}) : \mathbb{Q}\right] = p$ and $\left[\mathbb{Q}(\zeta_p) : \mathbb{Q}\right] = p-1$, so

$$\left[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}\right] = p(p-1).$$

Example. $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

The $K$ is a splitting field for $(x^2-2)\cdot(x^2-3)$




But, the splitting field of $x^3-2$ is not just $\mathbb{Q}(\sqrt[3]{2})$, because there are three roots;

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\cdot\left(\frac{-1+i\cdot\sqrt{3}}{2}\right), \quad \sqrt[3]{2}\cdot\left(\frac{-1-i\sqrt{3}}{2}\right)$$

Let $K$ be the splitting field of $x^3-2$. That is,

$$\sqrt[3]{2} \in K \quad \text{and} \quad \sqrt[3]{2}\cdot\left(\frac{-1+i\sqrt{3}}{2}\right) \in K,$$

so $\sqrt{-3} \in K$. It follows that $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.

Since $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) = \mathbb{Q}(\sqrt[3]{2})\,\mathbb{Q}(\sqrt{-3})$ and

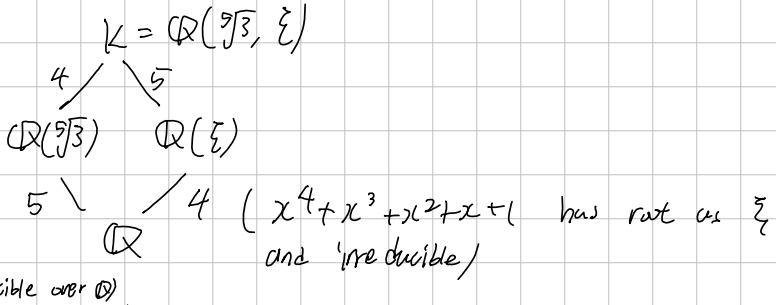
$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] = 2$, these are relatively prime, thus the degree of composite field is 6.

Find the Galois group of $x^5-3 \in \mathbb{Q}(\xi)[x]$ over $\mathbb{Q}(\xi)$,
where $\xi$ is a primitive fifth root of unity.

Proof. Let $K$ be the splitting field of $x^5-3$.
Since all roots of $x^5-3$ are

$$\sqrt[5]{3} \cdot \xi^k \quad (k=0,1,2,3,4) \quad \text{where } \xi = \exp\left(2\pi i \cdot \frac{1}{5}\right) = e^{\frac{2}{5}\pi i},$$

Therefore $K = \mathbb{Q}(\sqrt[5]{3}, \xi)$. Now,




($\because x^5-3$ is irreducible over $\mathbb{Q}$ ; $x^4+x^3+x^2+x+1$ has root as $\xi$ and irreducible)

Since $\mathbb{Q}(\sqrt[5]{3}, \xi) = \mathbb{Q}(\sqrt[5]{3}) \cdot \mathbb{Q}(\xi)$ and both degree over $\mathbb{Q}$ are relatively prime,
$[K:\mathbb{Q}] = 20$. And, since $[K:\mathbb{Q}(\xi)] = 5$, prime,

$$Gal(K/\mathbb{Q}(\xi)) \cong \mathbb{Z}_5.$$